

List rainbow connection number of graphs

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Abstract

An edge-coloured path is *rainbow* if all of its edges have distinct colours. Let G be a connected graph. The *rainbow connection number* of G , denoted by $rc(G)$, is the minimum number of colours in an edge-colouring of G such that, any two vertices are connected by a rainbow path. The *strong rainbow connection number* of G , denoted by $src(G)$, is the minimum number of colours in an edge-colouring of G such that, any two vertices are connected by a rainbow geodesic (i.e., a path of shortest length). These two notions of connectivity of graphs were introduced by Chartrand, Johns, McKeon and Zhang in 2008. In this paper, we introduce the *list rainbow connection number* $rc^\ell(G)$, and the *list strong rainbow connection number* $src^\ell(G)$. These two parameters are the versions of $rc(G)$ and $src(G)$ that involve list edge-colourings. Among our results, we will determine the list rainbow connection number and list strong rainbow connection number of some specific graphs. We will also characterise all pairs of positive integers a and b such that, there exists a connected graph G with $src(G) = a$ and $src^\ell(G) = b$, and similarly for the pair rc^ℓ and src^ℓ . Finally, we propose the question of whether or not we have $rc(G) = rc^\ell(G)$, for all connected graphs G .

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1 Introduction

In this paper, all graphs are finite, simple, and undirected. Let K_n and C_n denote the complete graph and cycle on n vertices. Let $K_{n_1, \dots, n_t}$ denote the complete t -partite graph with class sizes $n_1, \dots, n_t$. In particular, $K_{m,n}$ is the complete bipartite graph with class sizes m and n . For $t_1, t_2, \dots \geq 0$, let $K_{t_1 \times 1, t_2 \times 2, \dots}$ denote the complete multipartite graph with t_i classes of size i , for all $i \geq 1$. Any subscript with $t_i = 0$ is usually omitted. In particular, $K_{t \times n}$ denotes the complete t -partite graph where every class has size n . For a graph G ,

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